

CS2601 Linear and Convex Optimization

Homework 8

December 24, 2023

This assignment is not required to be submitted and will not be graded. You can check your answers by the solutions in the file CS2601hw8_sol.pdf.

1. Consider the following optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}} \quad & f(x) = \log(1 + e^x) \\ \text{s. t.} \quad & x \geq 0 \end{aligned}$$

- (a). Find the optimal solution and the optimal value.
- (b). Find the dual function and the dual problem.
- (c). Find the dual optimal solution and the dual optimal value. Does strong duality hold?

2. Consider the following problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & f(\mathbf{x}) = x_1^2 + x_2^2 \\ \text{s. t.} \quad & (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \\ & (x_1 - 1)^2 + (x_2 + 1)^2 \leq 1 \end{aligned}$$

- (a). Sketch the feasible set and the level set of the objective function. Find the optimal point $\mathbf{x}^*$ and the optimal value f^* .
- (b). Find the Lagrange dual function and the dual problem. Find the dual optimal value ϕ^* . Does strong duality hold?
- (c). Does Slater's condition hold? What can you conclude about the necessity of Slater's condition for strong duality?
- (d). Is the dual optimal value ϕ^* attained by any dual feasible point? Do there exist any Lagrange multipliers satisfying the KKT conditions at $\mathbf{x}^*$?

3. Consider the following minimization problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad f(\mathbf{x}) &= \begin{cases} x_1^3 + x_2^3, & \text{if } \mathbf{x} \geq \mathbf{0} \\ +\infty, & \text{otherwise} \end{cases} \\ \text{s.t.} \quad x_1 + x_2 &\geq 1 \end{aligned} \tag{P1}$$

Note the domain of f is $\text{dom } f = \{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x} \geq \mathbf{0}\}$ and the domain of the problem is $D = \text{dom } f$.

(a). Since D is not the entire space, the dual function of this problem is defined by

$$\phi(\mu) = \inf_{\mathbf{x} \in D} \{f(\mathbf{x}) + \mu(1 - x_1 - x_2)\}$$

Find the explicit expression of $\phi(\mu)$.

(b). Find the dual optimal solution.

(c). What is the primal optimal value? Hint: Note f is convex on its domain.

(d). Note the primal problem (P1) is equivalent to

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad f_1(\mathbf{x}) &= x_1^3 + x_2^3 \\ \text{s.t.} \quad x_1 + x_2 &\geq 1 \\ \mathbf{x} &\geq \mathbf{0} \end{aligned} \tag{P2}$$

What's the dual function of this equivalent problem (P2)? Does strong duality hold for (P2)?

Remark. Note $\text{dom } f_1 = \mathbb{R}^2$ and f_1 is not convex. This problem shows that equivalent primal problems can have very different dual problems. Not all dual problems are equally useful.